

An Introduction to Microsoft Graph

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About Me



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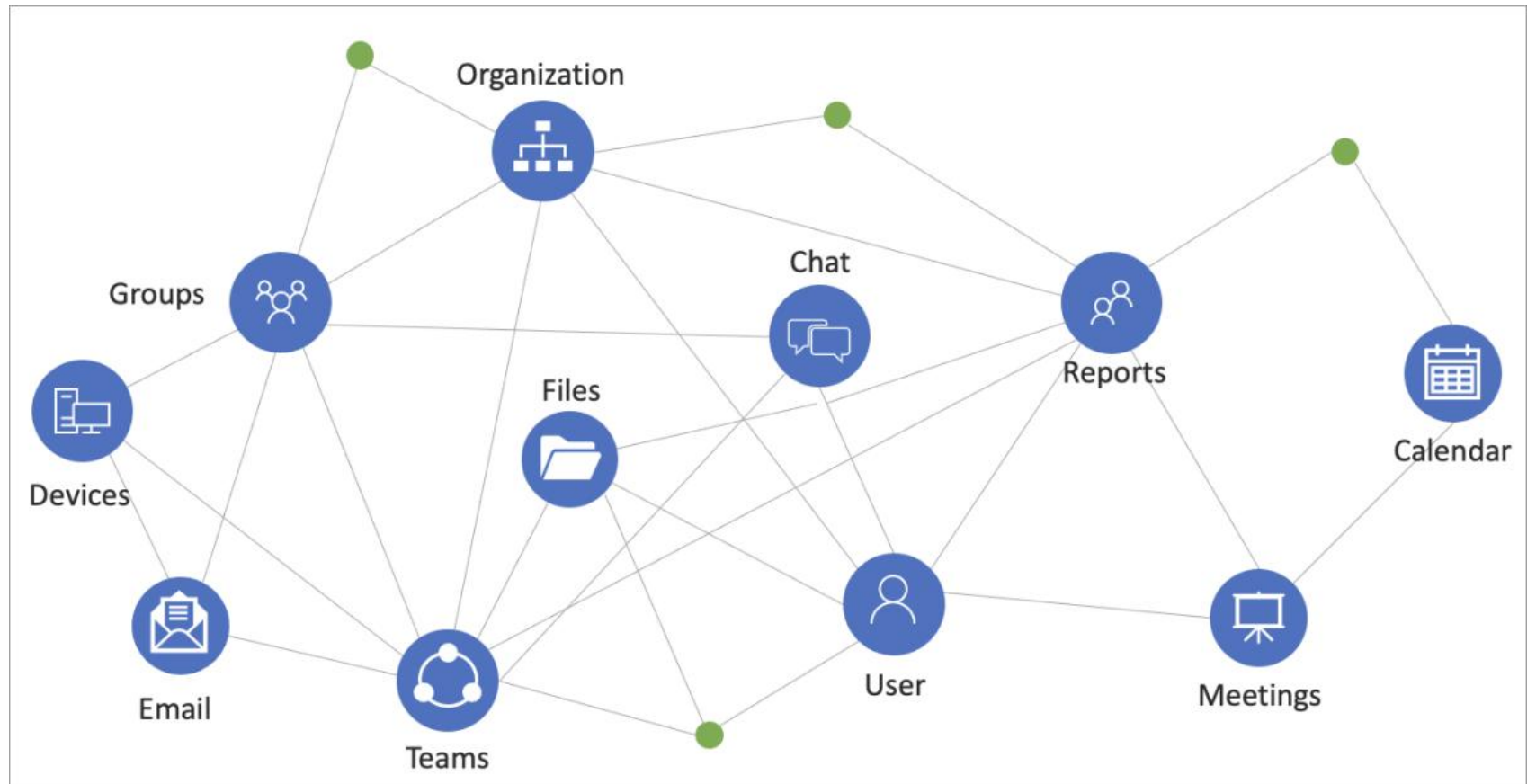
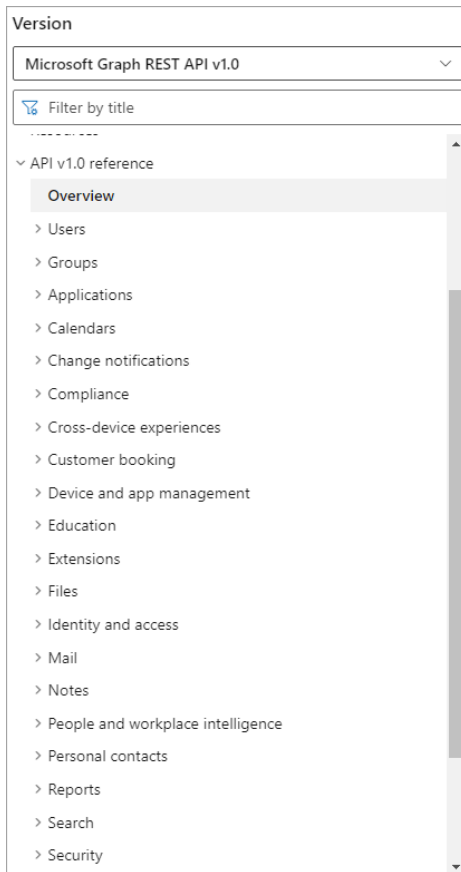
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Microsoft Graph

Unified API to connect to data related to various Microsoft 365 services



Tools and Resources

Microsoft Graph Developer Center

- graph.microsoft.com

Microsoft Graph Documentation

- learn.microsoft.com/en-us/graph

Microsoft Graph Tutorials

- learn.microsoft.com/en-us/graph/tutorials

Microsoft Graph Training

- developer.microsoft.com/en-us/graph/gallery/?filterBy=Training

Microsoft 365 developer tenant

- developer.microsoft.com/en-us/microsoft-365/dev-program

Graph Explorer

Graph Explorer

Tenant Sample ? ? ? ? ?

Sample queries Resources History

Search sample queries

See more queries in the [Microsoft Graph API Reference docs](#).

> Getting Started (8)

> Applications (8)

> Batching (2)

> Compliance (beta) (10)

> Excel (7)

> Extensions (7)

> Groups (13)

> Identity and Access (14)

GET v1.0 https://graph.microsoft.com/v1.0/me Run query

Request body Request headers Modify permissions Access token

OK - 200 - 1098ms

Response preview Response headers Code snippets Toolkit component Adaptive cards Expand

You are currently using a sample account. Sign in to access your own data.

```
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#users/$entity",
  "businessPhones": [
    "+1 412 555 0109"
  ],
  "displayName": "Megan Bowen",
  "givenName": "Megan",
  "jobTitle": "Auditor",
  "mail": "MeganB@M365x214355.onmicrosoft.com",
  "mobilePhone": null,
  "officeLocation": "12/1110",
  "preferredLanguage": "en-US",
  "surname": "Bowen",
}
```

Graph Application Basics

Register an application with Entra ID (Azure Active Directory)

Get an access token

Call the Graph

Entra ID Authentication

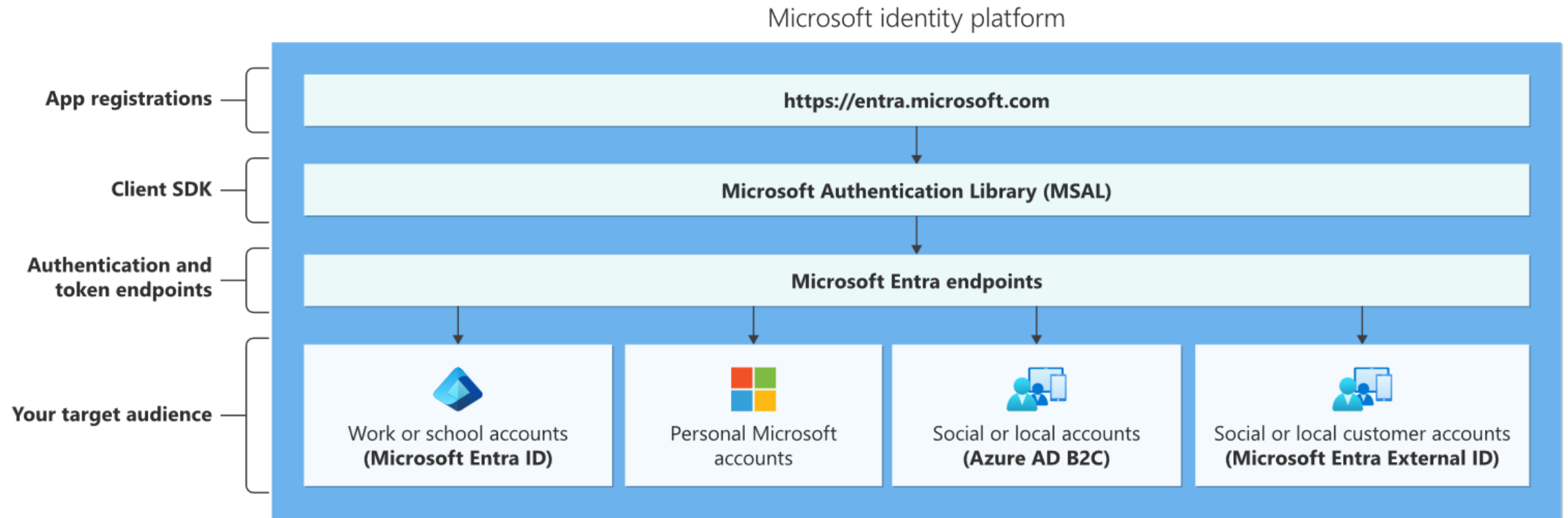
Calling the Graph is relatively easy

- It is not much different than calling any other RESTful API
- Not all endpoints support all OData query expressions

Understanding how to register an app and authenticate with Entra can be the difficult part

There are multiple APIs that can be used to assist with the authentication process and token exchange

Microsoft Identity Platform



Registering an Azure App

Azure Portal

- Based on OAuth version 2.0
- Supports both work and school accounts and personal accounts
 - Converged authentication

Static Permission Requests

Apps may make static permission requests

- Some permissions can be granted by user
- Some permissions can only be granted by an administrator

The screenshot shows the 'Request API permissions' dialog in the Azure Active Directory portal. The dialog is for the 'Microsoft Graph' API. It asks 'What type of permissions does your application require?' and offers two options: 'Delegated permissions' (selected) and 'Application permissions'. Below this, it says 'Select permissions' and shows a search bar with 'Mail.Read' entered. A table lists the selected permissions, with 'Mail (1)' expanded. The table has columns for 'Permission' and 'Admin consent required'. The 'Mail.Read' permission is selected, and its description is 'Read user mail'. The 'Admin consent required' column shows a checkmark for 'Mail.Read' and a minus sign for the others.

Dashboard > robwindortest986 | App registrations > InteractiveConsole | API permissions

InteractiveConsole | API permissions

Search (Ctrl+/) Refresh

Overview
Quickstart
Manage
Branding
Authentication
Certificates & secrets
Token configuration (preview)
API permissions
Expose an API
Owners
Roles and administrators (Previ...
Manifest
Support + Troubleshooting
Troubleshooting
New support request

Configured permissions

Applications are authorized to call APIs when they are granted per all the permissions the application needs. [Learn more about permi](#)

+ Add a permission Grant admin consent for robw

API / Permissions name	Type	Descripti
Microsoft Graph (1)		
User.Read	Delegated	Sign in an

Request API permissions

< All APIs

Microsoft Graph
<https://graph.microsoft.com/> Docs

What type of permissions does your application require?

Delegated permissions
Your application needs to access the API as the signed-in user.

Application permissions
Your application runs as a background service or daemon without a signed-in user.

Select permissions expand all

Mail.Read ✓

Permission	Admin consent required
☒ Mail (1)	
☒ Mail.Read Read user mail ⓘ	-
☐ Mail.Read.Shared Read user and shared mail ⓘ	-
☐ Mail.ReadBasic Read user basic mail ⓘ	-
☐ Mail.ReadWrite Read and write access to user mail ⓘ	-
☐ Mail.ReadWrite.Shared	-

Add permissions Discard

Dynamic Permission Requests

Only available when using delegated permissions

Permissions requests declared in code rather than in portal where application is registered

APIs and SDKs

Microsoft Authentication Library (MSAL)

Azure Identity Client Library

Microsoft Graph SDK

Tokens

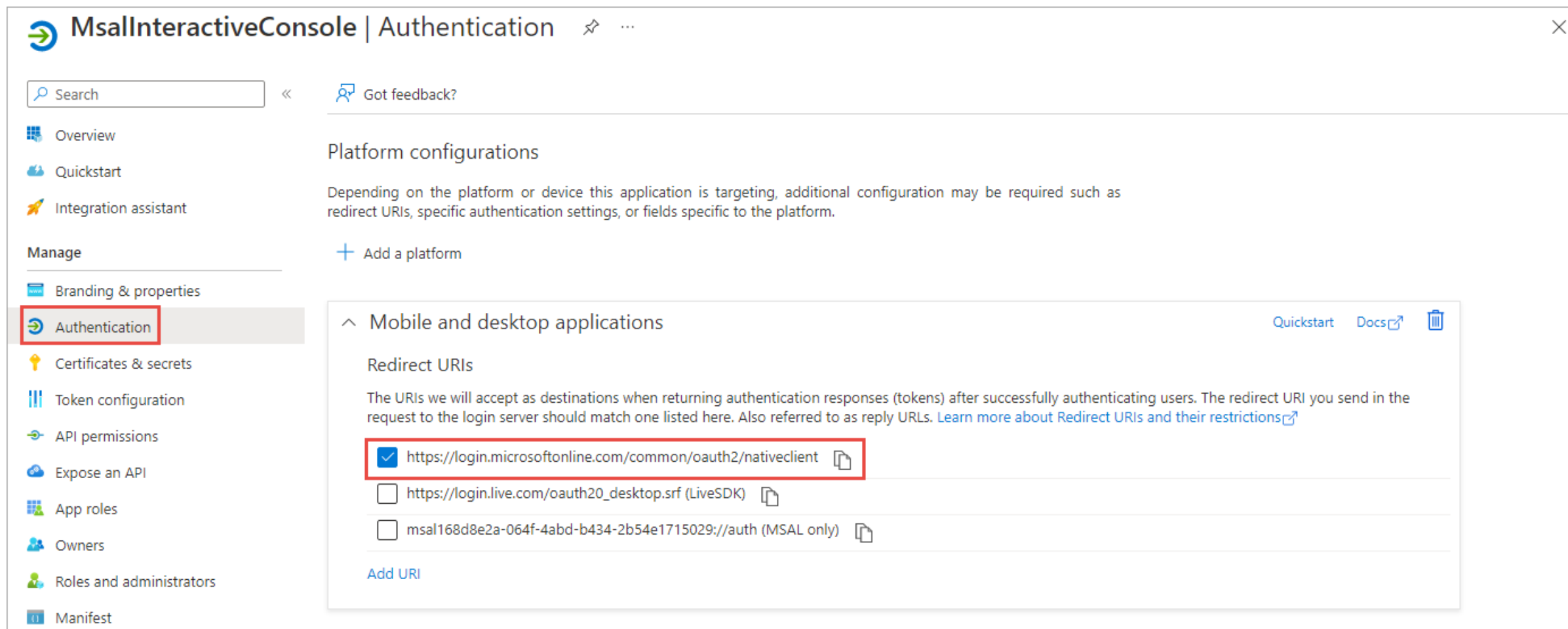
ID tokens

- Enable clients to get identity and profile information for a user
- Should not be used for authorization

Access and Refresh tokens

- Access tokens enable clients to securely call APIs protected by Azure
- Application can request an access token after successfully authenticating
 - May also be given a refresh token
- Access tokens are included in the headers of calls to the Graph
 - Graph uses token to confirm that:
 - You have authenticated
 - You have permission to access requested resources
 - Access tokens are valid for one hour
- Refresh tokens can be used to get a new access token after expiration

Interactive Flow App Registration - 2



The screenshot shows the MsalInteractiveConsole interface. The left sidebar contains a navigation menu with the following items: Overview, Quickstart, Integration assistant, Manage (with a sub-menu), Branding & properties, Authentication (highlighted with a red box), Certificates & secrets, Token configuration, API permissions, Expose an API, App roles, Owners, Roles and administrators, and Manifest. The main content area is titled 'Authentication' and includes a search bar, a 'Got feedback?' link, and a 'Platform configurations' section. The 'Platform configurations' section explains that additional configuration may be required based on the target platform and provides an 'Add a platform' button. Below this, the 'Mobile and desktop applications' section is expanded, showing a list of 'Redirect URIs'. The first URI, 'https://login.microsoftonline.com/common/oauth2/nativeclient', is selected with a checked checkbox and highlighted with a red box. The other two URIs are 'https://login.live.com/oauth20_desktop.srf (LiveSDK)' and 'msal168d8e2a-064f-4abd-b434-2b54e1715029://auth (MSAL only)'. A 'Quickstart' link, 'Docs' link, and a trash icon are visible in the top right of the 'Mobile and desktop applications' section. An 'Add URI' link is at the bottom of the list.

MsalInteractiveConsole | Authentication

Search << Got feedback?

Platform configurations

Depending on the platform or device this application is targeting, additional configuration may be required such as redirect URIs, specific authentication settings, or fields specific to the platform.

+ Add a platform

Mobile and desktop applications Quickstart Docs

Redirect URIs

The URIs we will accept as destinations when returning authentication responses (tokens) after successfully authenticating users. The redirect URI you send in the request to the login server should match one listed here. Also referred to as reply URLs. [Learn more about Redirect URIs and their restrictions](#)

- ☒ https://login.microsoftonline.com/common/oauth2/nativeclient
- ☐ https://login.live.com/oauth20_desktop.srf (LiveSDK)
- ☐ msal168d8e2a-064f-4abd-b434-2b54e1715029://auth (MSAL only)

Add URI

Interactive Flow Code (Desktop) - 1

```
private static async Task<string> GetAccessToken()
{
    var clientId = "43ec3caf-b22b-4fe3-84fd-340bd5b384cb";
    var authority = $"https://login.microsoftonline.com/{tenantName}.onmicrosoft.com/";
    var azureApp = PublicClientApplicationBuilder.Create(clientId)
        .WithAuthority(authority)
        .WithRedirectUri("http://localhost")
        .Build();

    var scopes = new string[] { "User.Read", "Mail.ReadBasic", "Files.Read" };
    var authResult = await azureApp.AcquireTokenInteractive(scopes).ExecuteAsync();

    return authResult.AccessToken;
}
```


Interactive Flow Code (Desktop) - 2

```
private async static Task MakeCallToMicrosoftGraph()
{
    var token = await GetAccessToken();
    var client = new HttpClient();
    client.DefaultRequestHeaders.Authorization = new AuthenticationHeaderValue("Bearer", token);
    client.DefaultRequestHeaders.Accept.Add(new MediaTypeWithQualityHeaderValue("application/json"));

    using (var response = await client.GetAsync($"https://graph.microsoft.com/v1.0/me/messages"))
    {
        response.EnsureSuccessStatusCode();

        var json = await response.Content.ReadAsStringAsync();
        var json = await response.Content.ReadAsStringAsync();
        var jsonObject = JsonObject.Parse(json);
        Console.WriteLine("Profile:");
        Console.WriteLine($"Your name is {jsonObject?["displayName"]}. Your title is {jsonObject?["jobTitle"]}");
    }
}
```

Interactive Flow Code (Desktop) (Graph SDK) - 1

```
public class TokenProvider : IAccessTokenProvider
{
    public async Task<string> GetAuthorizationTokenAsync(
        Uri uri,
        Dictionary<string, object> additionalAuthenticationContext = default,
        CancellationToken cancellationToken = default)
    {
        var token = await GetAccessToken();
        return token;
    }

    public AllowedHostsValidator AllowedHostsValidator { get; }
}
```

Interactive Flow Code (Desktop) (Graph SDK) - 2

```
private async static Task MakeCallToMicrosoftGraph()
{
    var provider = new BaseBearerTokenAuthenticationProvider(new TokenProvider());
    var client = new GraphServiceClient(provider);
    var messages = await client.Me.Messages.Request().GetAsync();
    foreach (var message in messages)
    {
        Console.WriteLine($"{message.Subject} from {message.Sender.EmailAddress}");
    }
}
```

SharePoint Framework Microsoft Graph Client

MSGraphClientV3

- @microsoft/sp-http package
- Abstracts token acquisition from SPFx support for Entra ID
- Wraps the Microsoft Graph JavaScript SDK

PnPjs

- @pnp/graph package
- Alternative to MSGraphClientV3 from the M365 Platform Community
- Includes fluent API to access Microsoft Graph

Can optionally use with Microsoft Graph Type Declarations

- @microsoft/microsoft-graph-types package
- Work with strongly-typed objects representing Graph data

SPFx Microsoft Graph Client Code

```
import { MSGraphClientV3 } from '@microsoft/sp-http';
import { GraphError } from '@microsoft/microsoft-graph-client';
import * as MicrosoftGraph from '@microsoft/microsoft-graph-types';

-----

export interface IGroupCollection {
  value: MicrosoftGraph.Group[];
}

-----

this.context.msGraphClientFactory
  .getClient('3')
  .then((client: MSGraphClientV3): void => {
    client.api('/groups')
      .orderBy('displayName')
      .get((error: GraphError, response: IGroupCollection, rawResponse?: any) => {
        let elem: Element = this.domElement.querySelector("#groupsContainer");
        elem.innerHTML = response.value.map((group) => {
          return `<span>${group.displayName}</span>`;
        }).join(`<br />`);
      });
  });
```

SPFx Solutions Declare Permission Requests

When using MSGraphClientV3 you do not need to register an Entra ID app

- Microsoft will do this for you

Your project declares the permissions it requires

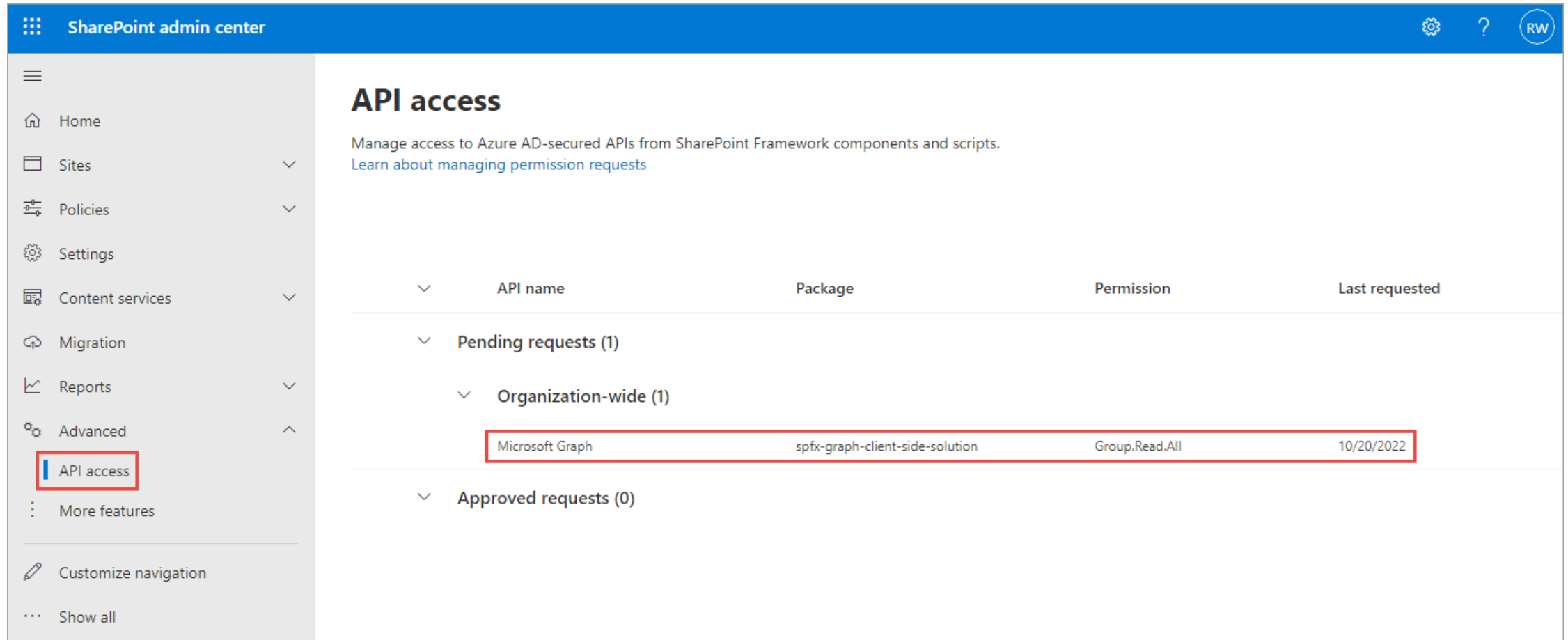
Tenant administrator grants or rejects permission requests using an API Management page in the SharePoint Online administration portal

- Granting the permission request updates the Entra ID app created by Microsoft

SPFx Solutions Declare Permission Requests

```
{
  "$schema": "https://developer.microsoft.com/json-schemas/spfx-build/package-solution.schema.json",
  "solution": {
    "name": "MSGraphClient Web Part",
    "id": "b9cbefa9-8399-42a3-923b-5261be6abf23",
    "version": "1.0.0.0",
    "includeClientSideAssets": true,
    "skipFeatureDeployment": true,
    "isDomainIsolated": false,
    "webApiPermissionRequests": [
      {
        "resource": "Microsoft Graph",
        "scope": "Group.Read.All"
      }
    ]
  },
  "paths": {
    "zippedPackage": "solution/ms-graph-client.sppkg"
  }
}
```

Approve / Reject with SharePoint Online API Management Page



The screenshot shows the SharePoint admin center interface. The left-hand navigation pane includes links for Home, Sites, Policies, Settings, Content services, Migration, Reports, Advanced, API access (highlighted with a red box), and More features. The main content area is titled "API access" and contains a table of pending requests. The table has columns for API name, Package, Permission, and Last requested. A single request is listed for "Microsoft Graph" with package "spfx-graph-client-side-solution", permission "Group.Read.All", and a last requested date of "10/20/2022". This row is highlighted with a red border. Below the pending requests, there is a section for "Approved requests (0)".

SharePoint admin center

API access

Manage access to Azure AD-secured APIs from SharePoint Framework components and scripts.
[Learn about managing permission requests](#)

API name	Package	Permission	Last requested
Pending requests (1)			
Organization-wide (1)			
Microsoft Graph	spfx-graph-client-side-solution	Group.Read.All	10/20/2022
Approved requests (0)			

Approve / Reject with SharePoint Online API Management Page

×

Approve access

If you approve access, any SharePoint Framework component or custom script can call this Azure AD-secured API with "Group.Read.All" permission.

API name Microsoft Graph	Package name spfx-graph-client-side-solution
Permission Group.Read.All	Version 1.0.0.0
Requested by Rob Windsor	Last requested 10/20/2022

Approve **Cancel**

Microsoft Learn Training

Microsoft Graph Fundamentals

- learn.microsoft.com/en-us/training/paths/m365-msgraph-fundamentals/

Explore Microsoft Graph scenarios for ASP.NET Core development

- learn.microsoft.com/en-us/training/paths/m365-msgraph-dotnet-core-scenarios/

Explore Microsoft Graph scenarios for JavaScript development

- learn.microsoft.com/en-us/training/paths/m365-msgraph-scenarios/

SPFx - Use Microsoft Graph and non-Microsoft APIs

- learn.microsoft.com/en-us/training/modules/sharepoint-spfx-graph-3rd-party-apis/